

Claude designs 1320 protein binders with open source software

Anthropic's Claude autonomously designed 1320 protein binders using only open source, freely downloadable design software, with Anthropic paying no software licensing fees. The technical report says AlphaFold 3 weights and two other packages were excluded because of restrictive licences.

Given 16 protein targets, cloud compute and a roughly 16,000-word protocol, Claude researched each target, selected binding sites, ran the software and produced 30 ranked designs per target. Two independent laboratories then synthesised and tested the candidates.

Of the 1320 designs, 354 bound their targets, producing a 27 per cent hit rate, while 49 per cent of Claude's top-ranked designs succeeded. At least one design worked for 14 of the 15 targets with readable measurements. Against RBX1, Claude produced 28 binders from 90 designs, compared with nine from 245 entries in a human open design competition. Its strongest design bound roughly 10 times more tightly than the competition winner in side-by-side testing.

However, the experiment also exposed the next bottleneck: physical synthesis and experimental screening. Adaptyv Bio tested all 1320 designs, while Twist Bioscience tested 1260 independently. Experimental screening remains the only way to learn which targets a campaign has succeeded on and the affinity of the designed binders.

Anthropic has published the prompts and data on Hugging Face for others to rerun. The study demonstrated binding, not biological function, and does not establish that Claude outperformed human experts.

The open framework is intended to connect data, models, software and different robotic platforms on a common foundation. This could allow robots to continuously learn, adapt and improve through real-world deployment while giving the wider community an environment for developing and sharing embodied intelligence.

The addition of AIRSEAI strengthens LF AI & Data Foundation's efforts to establish open and interoperable infrastructure for embodied AI. By bringing the project into a neutral open source ecosystem, the foundation aims to reduce proprietary barriers and support collaboration around robotics systems that can operate across different hardware architectures and deployment environments.

Audacity 4.0 brings a new look to audio editing

The open source audio editor Audacity 4.0 has officially been released, marking one of the biggest updates in the project's history. The new version rebuilds the application's interface using Qt and introduces a redesigned editing experience, more flexible recording controls and major changes to how projects and audio clips are handled.



One of the biggest changes is the new clip-editing model. Users can directly select individual clips or multiple clips and edit them together. Clips can also be grouped so they remain together when moved, copied, or duplicated. The update further allows clips to move more freely between compatible mono and stereo tracks, while new splitting tools, alignment guides and sample-boundary snapping expand editing options.

The interface has also received a complete overhaul with native high-DPI rendering. Toolbars and panels can now be moved, docked, floated, shown or hidden, while users can save different layouts as 'workspaces'. Audacity 4.0 includes 'Modern', 'Classic' and 'Music' workspaces, alongside light, dark and high-contrast themes, accent colours, track colours, and different clip styles.

Audacity 4.0 also changes several core workflows. Recording can begin from anywhere on the timeline, while playback can seek to another position without stopping. Punch and roll, lead-in recording, latency compensation, software playthrough and per-track input monitoring have also been redesigned. The older 'Select', 'Envelope', 'Draw', and 'Multi-tool' modes have been replaced with context-sensitive controls.

However, some Audacity 3 features, including 'Time Tracks', 'Note/MIDI tracks', 'Mixer', 'Macro Manager', 'Scripting pipe', 'VAMP' and 'LADSPA' plugin hosting and 'Play-at-Speed' are not yet available in version 4.0.